# ShorpSmart Retail Ltd

## Understanding Data

### 1. Number of Datasets Provided

10 datasets were provided with 9 core business tables that included Customers, Products, Orders, Order items, Stores, Returns, Employees, Suppliers and Inventory including Data Dictionary.

### 2. Description of Datasets

| Table | Rows | Description |
|---|---|---|
| **Customers** | 5,000 | Incudes information of customers demography, contact details, registration date, segments (Bronze/Silver/Gold). |
| **Products** | 500 | Includes products names, categories, brand, cost price, selling price of products. |
| **Orders** | 50,000 | Includes information of customers, products, store, date, quantity, price, discount, shipping, payments etc. on orders made. |
| **Order Items** | 50,000 | Includes information quantity, unit price, discount, line total of each item that was ordered. |
| **Stores** | 25 | Includes name, region, state, city, opening date describing locations of different stores across Nigeria. |
| **Returns** | 2,500 | Orders that were returned showing information like return date, reason, refund amount. |
| **Employees** | 125 | Staff records including names, roles, stores assigned, date they were hired, monthly salary. |
| **Suppliers** | 489 | These include information about suppliers showing name, email, phone, city, country. |
| **Inventory** | 500 | Stock levels by products & store. |
| **Data Dictionary** | 41 | Description of the 9 datasets. |

### 3. Primary Keys and Foreign Keys

| Table | Primary Keys | Foreign Key(s) | References |
|---|---|---|---|
| **Customers** | Customers_ID | - | - |
| **Products** | Products_ID | - | - |
| **Orders** | Orders_ID | Customer_ID, Product_ID, Store_ID | Customer, Product, Store |
| **Order Items** | Order Items_ID | Order_ID, Product_ID | Orders, Products |
| **Stores** | Stores_ID | - | - |
| **Returns** | Returns_ID | Order_ID | Orders |
| **Employees** | Employees_ID | Store_ID | Stores |
| **Suppliers** | Suppliers_ID | - | - |
| **Inventory** | Inventory_ID | Product_ID, Store_ID | Product, Stores |

### 4. Relationships between Tables

The ShopSmart datasets contain several one-to-many relationships established through primary and foreign keys. These relationships connect different datasets and allow information to be analysed across multiple tables. For example, one customer can place multiple orders, but each order is associated with a particular customer. Similarly, one product can appear in multiple orders and order items because the same product can be purchased several times. The Stores table is related to the Orders, Employees and Inventory tables through Store_ID. This allows each store to be associated with multiple sales transactions, employees and inventory records. The Orders table is also connected to the Order Items and Returns tables through Order_ID, while the Products and Stores tables connect to Inventory through Product_ID and Store_ID respectively. Only Suppliers tables does not show any connection with other tables demonstrating that it does not have any relation in the datasets.